

P1468R4

Fixed-layout floating-point type aliases

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This version:

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§ 1. Why this paper is empty

This paper has been merged into [P1467](#), starting with [P1467R4](#). P1467 (extended floating-point types) and P1468 (standard names for well-known layouts) were separate proposals when first written. But over time they have become intertwined, with design decisions in one paper affecting the feasibility of the other. So the two papers have now been merged into a single proposal. This version of P1468 exists merely to point people to where the contents now live.

§ 2. Revision history

§ 2.1. R0 -> R1 (pre-Cologne)

1. Add the requirement that the types must not alias any of the standard floating-point types.
2. Add a design question about feature-test macros.
3. Add a section on QoI - should we strongly encourage that the aliases to have a hardware implementation?

§ 2.2. R1 -> R2 (pre-Belfast)

Changes based on feedback in Cologne from [SG6](#), [LEWGI](#), and [EWGI](#). Further changes came from further development of the paper by the authors.

- Expanded the [section](#) about whether or not the fixed-layout aliases are allowed to alias standard floating-point types.
- Added a [section](#) about whether the aliases only need to guarantee layout, or should also guarantee behavior.
- Added some text, still preliminary, about [literal suffixes](#).

§ 2.3. R2 -> R3 (pre-Prague)

Changes based on feedback in Belfast from [EWG](#).

- Added `<stdfloat>` as a possible name for the header containing the proposed type aliases.
- Resolved the issue of [layout vs. behavior](#) by stating that `std::numeric_limits<T>::is_iec559` is true for the IEEE aliases.
- Reduced the proposed sets of names for the type alias from six different naming schemes down to three.

§ 2.4. R3 -> R4 (Summer 2020)

Merge the contents into [P1467R4](#), leaving an empty shell of a paper.